Alternating multilinear maps and wedge product

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We will be working over $\mathbf{Vect}_k$ for some arbitrary field k of characteristic 0.

Using the machinery developed in the previous section about groups acting on vector spaces, we have that:

- Consider the action of S_n on $T^n(V)$ by

$$\sigma(v_1 \otimes ... \otimes v_n) = \text{sgn}(\sigma)\Big(v_{\sigma^{-1}(1)} \otimes ... \otimes v_{\sigma^{-1}(n)}\Big).$$

  Then, linear maps from $T^n(V)$ to W (i.e., multilinear maps from $(V, ..., V)$ to W) are called "alternating" if they respect this action; i.e.,

$$\varphi \in \text{Hom}_k(T^n(V), W) \text{ is called alternating iff}$$
$$\forall v \in T^n(V), \forall \sigma \in S_n, \varphi(\sigma(v)) = \varphi(v).$$

- Define the nth wedge product of V as

$$\Lambda^n(V) := \frac{T^n(V)}{\text{span}_k(\{v - w : v \in V, w \in \text{orbit}(v)\})}$$

  $\Lambda^n(V)$ has the nice property that alternating multilinear maps from $(V, ..., V)$ to W are in one-to-one correspondence with linear maps from $\Lambda^n(V)$ to W. So, we have

$$\text{AltMulti}(V, ..., V; W) \cong \text{Hom}_k(\Lambda^n(V), W)$$

- We write $v_1 \wedge ... \wedge v_n \in \Lambda^n(V)$ for the image of $v_1 \otimes ... \otimes v_n \in T^n(V)$ under the quotient map.
- A vector in $T^n(V)$ is called an "alternating tensor" if it is stable under the action of S_n; i.e., it has a singleton orbit; i.e.,

$$\forall \sigma \in S_n, \sigma(v) = v.$$

  We shall denote the set of alternating tensors in $T^n(V)$ as V^n_{alt}.

- Consider the map

$$\text{Alt}^n \in \text{End}_k(T^n(V))$$
$$\forall v \in T^n(V), \text{Alt}^n(v) := \frac{1}{n!} \sum_{\sigma \in S_n} \sigma(v).$$

  Note that in the above definition, $\sigma(v) = \sigma(v_1 \otimes ... \otimes v_n) = \text{sgn}(\sigma) \times v_{\sigma^{-1}(1)} \otimes ... v_{\sigma^{-1}(n)}$, since that is how σ acts on v.

  It is easy to show that $\text{im}(\text{Alt}^n) = V^n_{\text{alt}}$.

  This gives us a characterization of how the alternating tensors *look* like.

Our definition of Alt^n boils down to

$$v_1 \otimes ... \otimes v_n \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \Big(v_{\sigma^{-1}(1)} \otimes ... \otimes v_{\sigma^{-1}(n)} \Big)$$

Note that some authors define Alt^n as

$$v_1 \otimes ... \otimes v_n \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \Big(v_{\sigma(1)} \otimes ... \otimes v_{\sigma(n)} \Big),$$

but don't worry, since we are taking sum over the entirety of S_n, so σ and σ^{-1} come in pairs; and $\text{sgn}(\sigma) = \text{sgn}(\sigma^{-1})$. Both definitions are equivalent.

- Alt^n is a projection operator on $T^n(V)$ (projecting onto V^n_{alt}), and we have

  $$\ker(\text{Alt}^n) = \text{span}_k(\{v - w : v \in V, w \in \text{orbit}(v)\}).$$

  Therefore,

  $$\Lambda^n(V) = \frac{T^n(V)}{\ker(\text{Alt}^n)} \cong \text{im}(\text{Alt}^n) = V^n_{\text{alt}},$$

  using the first isomorphism theorem.

  Explicitly writing the isomorphism out, we have

  $$\left([v] \mapsto \text{Alt}^n(v) = \frac{1}{n!} \sum_{\sigma \in S_n} \sigma(v) \right) : \Lambda^n(V) \to V^n_{\text{alt}}$$
  $$(v \mapsto [v]) : V^n_{\text{alt}} \to \Lambda^n(V)$$

  Again, note that $\sigma(v) = \sigma(v_1 \otimes ... \otimes v_n) = \text{sgn}(\sigma) \times v_{\sigma^{-1}(1)} \otimes ... v_{\sigma^{-1}(n)}$, since that is how σ acts on v.

  Look at the section below where we try to define multiplication on $\oplus_{n \geq 0} V^n_{\text{alt}}$.

  Spivak might **not** be using this nice canonical isomorphism, since that will not match with their definition of

  $$v \wedge w = \binom{k+l}{k} \text{Alt}^{k+l}(v \otimes w)$$

  Spivak uses the same definition of Alt^n that we do, so that cannot be the difference.

  If instead of using the isomorphism $v \mapsto [v]; [v] \mapsto \text{Alt}^n(v)$ between $\Lambda^n(V)$ and V^n_{alt}, for some reason they have chosen to use the non-canonical definition,

  $$v \mapsto \frac{1}{n!}[v]$$
  $$[v] \mapsto n! \text{Alt}^n(v) = \sum_{\sigma \in S_n} \sigma(v),$$

  then they will indeed get the above multiplication factor of $\binom{k+l}{k}$.

  So ultimately, what we get is:

- The canonical isomorphism described above doesn't give us the $\binom{k+l}{k}$ factor in multiplication, and for some reason Spivak wants that factor.
- That factor arises if instead of having the canonical isomorphism $[v] \mapsto \text{Alt}^n(v)$, we instead have $[v] \mapsto n!\text{Alt}^n(v)$.
- Spivak may have chosen that because $n!\text{Alt}^n(v) = \sum_{\sigma \in S_n} \sigma(v)$ instead of having $\frac{1}{n!} \sum_{\sigma \in S_n} \sigma(v)$.

Suppose V is a finite dimensional vector space with $\dim(V) = d$.

If $n > d$, then the subspace we are quotienting by in the expression for $\Lambda^n(V)$ is the entire space $T^n(V)$, and therefore $\dim(\Lambda^n V) = 0$.

If $n \leq d$, and V has a basis $\{v_1, ..., v_d\}$, then $\left\{v_{i_1} \wedge ... \wedge v_{i_n} : 1 \leq i_1 < i_2 < ... < i_n \leq d\right\}$ is a basis for $\Lambda^n(V)$. So,

$$n \leq d \Rightarrow \dim(\Lambda^n(V)) = \binom{\dim(V)}{n}.$$

Particularly, $\Lambda^{\dim(V)}(V)$ is isomorphic to the scalar field k.

Often we are interested in multilinear maps from $(V, ..., V)$ back to the base field k. This corresponds to

$$\text{Hom}_k(\Lambda^n(V), k) = (\Lambda^n(V))^*$$

Recall that, if V is a finite dimensional vector space, then we have

$$T^n(V^*) \cong (T^n(V))^*$$

The quotient map $\pi_1 : T^n(V) \twoheadrightarrow \Lambda^n(V)$ is a surjection, which induces an inclusion $\pi_1^* : \Lambda^n(V)^* \hookrightarrow T^n(V)^*$, and $(T^n(V))^* \cong T^n(V^*)$, which gives us the inclusion

$$\Lambda^n(V)^* \hookrightarrow T^n(V^*)$$

Again, assume V is finite dimensional.

Observe the following chain of morphisms:

$$\Lambda^n(V)^* \overset{\pi_1^*}{\hookrightarrow} T^n(V)^* \overset{i}{\to} T^n(V^*) \overset{\pi_2}{\twoheadrightarrow} \Lambda^n(V^*),$$

where $i : T^n(V)^* \overset{\sim}{\to} T^n(V^*)$ is from the canonical isomorphism $T^n(V)^* \cong T^n(V^*)$.

This chain of morphisms composes to the following map from $\Lambda^n(V)^*$ to $\Lambda^n(V^*)$:

$$f \mapsto \sum_{1 \leq i_1 < ... < i_n \leq d} f\left(e_{i_1} \wedge ... \wedge e_{i_n}\right) \cdot \left(e^{i_1} \wedge ... \wedge e^{i_n}\right) = \sum_{|I|=n} f(e_I) e^I,$$

where $\{e_i\}_{1 \leq i \leq d}$ is a basis for V and $\{e^i\}_{1 \leq i \leq d}$ is the dual basis.

We would like to show that the above chain is indeed an isomorphism between $\Lambda^n(V)^*$ and $\Lambda^n(V^*)$. Following are 2 approaches that might work.

In the category k-Vect, if we have an injection f followed by a surjection π in the following way

$$A \overset{f}{\hookrightarrow} B \overset{\pi}{\twoheadrightarrow} C,$$

then $\pi \circ f$ is an isomorphism iff $B \cong \mathrm{im}(i) \oplus \ker(\pi)$.

This is probably true for any Abelian category, need to check.

If we can show that $\mathrm{im}(\pi_1^*) \oplus i^{-1}(\ker(\pi_2)) \cong T^n(V)^*$, then we are done in showing that $\pi_2 \circ i \circ \pi_1^*$ is an isomorphism. Try again later. : (i) Decompose arbitrary $v \in T^n(V)^*$ into the direct sum components, and (ii) show that the direct sum components intersect at $\{0\}$.

If we can show that the chain of morphisms given above is injective, then equality of dimension will establish that it is a bijection.

It turns out that the above is indeed an isomorphism, and the inverse of this isomorphism is

$$\left[f_1 \wedge \ldots \wedge f_n \mapsto \left[u_1 \wedge \ldots \wedge u_n \mapsto \det\left[f_i(u_j)\right]\right]\right] : \Lambda^n(V^*) \xrightarrow{\sim} \Lambda^n(V)^*$$

Try again later. : Why is that so?

The proof of the above fact may require the following:

The Leibniz determinant formula states that:

$$\det\left(\left[a_{ij}\right]_{n\times n}\right) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) a_{1\sigma(1)} \ldots a_{n\sigma(n)}$$

Now, we would like to describe multiplication over $\oplus_{n\geq 0} \Lambda^n(V)$ and over $\oplus_{n\geq 0} V_{\mathrm{alt}}^n$.

For both multiplications, we would be using the same symbol "$\wedge$".

Recall that $\Lambda^n(V)$ is defined to be a quotient of $T^n(V)$, and we have already defined multiplication over $\oplus_{n\geq 0} T^n(V)$, so for $[v_1] \in \Lambda^k(V), [v_2] \in \Lambda^l(V)$ (for some $v_1 \in T^k(V), v_2 \in T^l(V)$), we define the product of $[v_1]$ and $[v_2]$ as

$$[v_1] \wedge [v_2] = [v_1 \otimes v_2] \in \Lambda^{k+l}(V)$$

Now, we can use the isomorphism $\Lambda^n(V) \cong V_{\mathrm{alt}}^n$ to get a multiplication on $\oplus_{n\geq 0} V_{\mathrm{alt}}^n$.

Now, things start going dirty, if we try to match the "standard convention".

The isomorphism that we gained canonically from the fact that the Reynold's operator (Alt^n) is a projection was

$$[v] \in \Lambda^n(V) \mapsto \mathrm{Alt}^n(v) \in V_{\mathrm{alt}}^n$$
$$v \in V_{\mathrm{alt}}^n \mapsto [v] \in \Lambda^n(V)$$

Let us try using this isomorphism to get multiplication on $\oplus_{n\geq 0} V_{\mathrm{alt}}^n$:

Given $v \in V_{\text{alt}}^k, w \in V_{\text{alt}}^l$, they correspond to $[v] \in \Lambda^k(V), [w] \in \Lambda^l(V)$, which multiply to give $[v \otimes w] \in \Lambda^{k+l}(V)$, which corresponds to $\text{Alt}^{k+l}(v \otimes w) = \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \sigma(v \otimes w)$. So, the multiplication we get is

$$v \wedge w = \text{Alt}^{k+l}(v \otimes w)$$

The above formulation doesn't have any $\binom{n+k}{k}$ scalar in it, while Spivak does. My guess at how that might be arising is that instead of using the above isomorphism, Spivak uses $v \mapsto \frac{1}{n!}[v]; [v] \mapsto n!\text{Alt}_n(v)$. Further details are described above where we discuss $\Lambda^n(V) \cong V_{\text{alt}}^n$. If we use *that* isomorphism, then

> Given $v \in V_{\text{alt}}^k, w \in V_{\text{alt}}^l$, they correspond to $\frac{1}{k!}[v] \in \Lambda^k(V), \frac{1}{l!}[w] \in \Lambda^l(V)$, which multiply to give $\frac{1}{k!l!}[v \otimes w] \in \Lambda^{k+l}(V)$, which corresponds to $\binom{k+l}{k}\text{Alt}^{k+l}(v \otimes w)$. So, the multiplication we get is
>
> $$v \wedge w = \binom{k+l}{k}\text{Alt}^{k+l}(v \otimes w)$$

Given basis $\{e_1, ..., e_d\}$ of V, and vectors $v_1, ..., v_d \in V$ such that

$$[e_1 \ \ ... \ \ e_d]A_{d \times d} = [v_1 \ \ ... \ \ v_d],$$

we have $v_1 \wedge ... \wedge v_d \in \Lambda^d(V)$, and simplifying, we get that

$$v_1 \wedge ... \wedge v_d = \det(A) \cdot e_1 \wedge ... \wedge e_d$$

Given a k-linear map $\varphi : V \to W$, we have a map $T^n(\varphi) : T^n(V) \to T^n(W)$ given by $(v_1 \otimes ... \otimes v_n) \mapsto (\varphi(v_1) \otimes ... \otimes \varphi(v_n))$.

Now, $\pi \circ T^n(\varphi) := (v_1 \otimes ... \otimes v_n) \mapsto \varphi(v_1) \wedge ... \wedge \varphi(v_n)$ is an alternating map from $T^n(V)$ to $\Lambda^n(W)$, and so we get an unique map $\Lambda^n(\varphi) : \Lambda^n(V) \to \Lambda^n(W)$.

> This shows that Λ^n is functorial, just like T^n.

Given basis $B_V = \{u_1, ..., u_k\}$ for V and $B_W = \{w_1, ..., w_l\}$ for W. Suppose the matrix of φ with respect to this basis is $A_{l \times k}$.

Suppose $n \leq \min(k, l)$. Then we have basis $B_V^0 = \left\{u_{i_1} \wedge ... \wedge u_{i_n} : 1 \leq i_1 < ... < i_n \leq k\right\}$ for $\Lambda^n(V)$ and $B_W^0 = \left\{w_{j_1} \wedge ... \wedge w_{j_n} : 1 \leq j_1 < ... < j_n \leq l\right\}$ for $\Lambda^n(W)$. Then, for some fixed $1 \leq i_1 < ... < i_n \leq k$, we have

$$\Lambda^n(\varphi)\left(u_{i_1} \wedge ... \wedge u_{i_n}\right) = \varphi(u_1) \wedge ... \wedge \varphi(u_n)$$

Fix some $1 \leq j_1 < ... < j_n \leq l$. We can write $\Lambda^n(\varphi)\left(u_{i_1} \wedge ... \wedge u_{i_n}\right)$ in the basis B_W^0, and the coordinate along $w_{j_1} \wedge ... \wedge w_{j_n}$ will be

$$
\begin{aligned}
&\sum_{\sigma\in S_n}\left(\prod w^{j_{\sigma(k)}}\left(\varphi\left(u_{i_k}\right)\right)\right)\cdot w_{j_{\sigma(1)}}\wedge\ldots\wedge w_{j_{\sigma(n)}}\\
&=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\cdot\left(\prod A_{\sigma(k),k}\right)\cdot w_{j_1}\wedge\ldots\wedge w_{j_n}\\
&=\det(\operatorname{minor}_{n\times n}(A,(i_1,\ldots,i_n),(j_1,\ldots,j_n)))\cdot w_{j_1}\wedge\ldots\wedge w_{j_n}
\end{aligned}
$$

So, the matrix of $\Lambda^n(\varphi)$ will have determinants of $n\times n$ minors of A as its entries.

Therefore, given $\varphi: V\to V$ with matrix A wrt basis $\{e_1,\ldots,e_d\}$, $\Lambda^d\varphi$ will have matrix $[\det(A)]$ with respect to the basis $\{e_1\wedge\ldots\wedge e_d\}$.

But now, a nice thing happens. $\Lambda^d(V)$ is 1-dimensional, and therefore the only data that $\Lambda^d(\varphi)$ stores is a single scalar value, which in this case happens to be $\det(A)$.

So, given $\varphi\in\operatorname{End}_k(V)$, $\Lambda^d(\varphi)\in\operatorname{End}_k\left(\Lambda^d(V)\right)$ is the map "scaling by $\det(\varphi)$".

This gives us a basis free definition of $\det(\varphi)$, so we can take **determinants of endomorphisms over finite dimensional vector spaces.**

The following discussion is when the base field k is equal to $\mathbb{R}$.

When do two basis $B_u=\{u_1,\ldots,u_d\}$ and $B_v=\{v_1,\ldots,v_d\}$ of a vector space V have the **same orientation**?

1st try When the change of basis matrix from B_u to B_v has positive determinant.

> This is a bad definition, as need to take determinant of matrix. Furthermore, asymmetric; why should we go from B_v to B_u and not from B_u to B_v?

2nd try $u_1\wedge\ldots\wedge u_d$ and $v_1\wedge\ldots\wedge v_d$ are both vectors in the 1-dimensional vector space $\Lambda^d(V)$, so each of them is a scalar multiple of the other. If the scalar is positive, then they have same orientation; otherwise not.

3rd try For every $\omega\in\Lambda^d(V)^*$, we have $\omega(u_1\wedge\ldots\wedge u_d)=\det(A)\omega(v_1\wedge\ldots\wedge v_d)$, if $[v_1\ \ldots\ v_d]A=[u_1\ \ldots\ u_d]$. Define that B_u and B_v have same orientation iff

$$
\forall\omega\in\Lambda^d(V)^*,\omega(u_1\wedge\ldots\wedge u_d)\text{ and }\omega(v_1\wedge\ldots\wedge v_d)\text{ have the same sign.}
$$
